

# FloodEye

## AWS Production Scaling & Infrastructure Cost Plan

### Dual Currency Analysis: USD (\$) & INR (INR @ 84.50/\$)

Technology Innovation Hub (TIH), IIT Guwahati | July 2025 | Region: AWS ap-south-1  
(Mumbai)

## 1 Executive Summary

This document outlines the production-ready infrastructure roadmap and dual-currency cost model (**USD** and **INR**) for transitioning the **FloodEye** IoT flood monitoring platform from a prototype to an enterprise-grade disaster warning system on AWS. The current prototype relies on HTTP polling via ThingSpeak and client-side inference. To support high-concurrency monsoon alerts across Assam and Northeast India, five core enhancement pillars are mapped to managed AWS services:

- 1. MQTT Protocol:** Migrating to mutual TLS MQTT ingestion via **AWS IoT Core**.
- 2. PostgreSQL Database:** Replacing JSON files with **Amazon RDS for PostgreSQL + TimescaleDB**.
- 3. Multi-Node Mesh Architecture:** Distributed containerized API execution on **AWS ECS Fargate + ALB**.
- 4. SMS/WhatsApp Gateway:** TRAI-compliant emergency alerting via **Amazon SNS, Lambda, and Twilio**.
- 5. Advanced ML Inference:** Server-side multi-horizon TFT forecasting on **Amazon SageMaker Serverless**.

All cost estimates assume a baseline **50-node regional deployment** transmitting every 15 seconds ( $\approx 8.64$  million telemetry messages/month) in **AWS ap-south-1 (Mumbai)**, benchmarked at an exchange rate of **\$1 USD = INR 84.50**.

## 2 AWS Service Architecture

### 2.1 Architecture Diagram

### 2.2 Pillar-by-Pillar Technical Specifications

- Pillar 1 – MQTT (AWS IoT Core):** Replaces HTTP polling with lightweight MQTT over TLS (Port 8883) using X.509 client certificates per node. Handles  $50 \times 4 \text{ msg/min} \times 43,200 \text{ min/mo} = 8.64 \text{ M}$  messages/month with QoS 1 reliability.
- Pillar 2 – PostgreSQL (Amazon RDS + TimescaleDB):** Managed `db.t4g.small` instance (2 vCPU, 2 GB RAM, 50 GB gp3 SSD) running PostgreSQL with the TimescaleDB hypertable extension for high-speed time-series queries and 7-day automated backups.

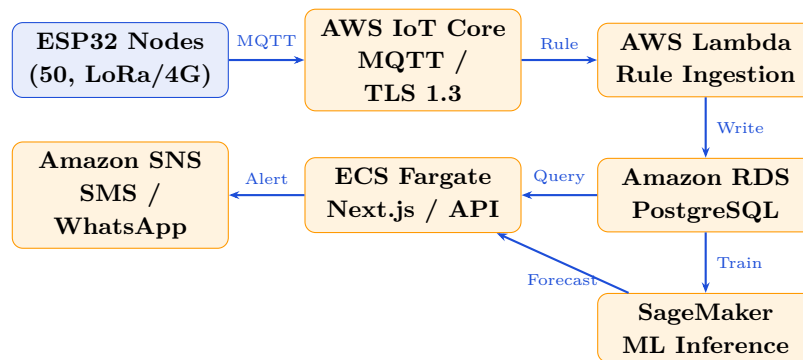


Figure 1: FloodEye Production Cloud Architecture on AWS ap-south-1 (Mumbai)

- **Pillar 3 – Multi-Node App (ECS Fargate + CloudFront):** Containerized Next.js frontend and API running across 2 Fargate tasks (0.5 vCPU, 1 GB RAM each) behind an Application Load Balancer. CloudFront CDN caches static assets, reducing network egress by 80%.
- **Pillar 4 – SMS Alert Gateway (SNS + Lambda + Twilio):** Event-driven threshold breaches invoke AWS Lambda to broadcast TRAI-compliant SMS and WhatsApp alerts. ElastiCache Redis (t4g.micro) deduplicates alerts within a 15-minute sliding window.
- **Pillar 5 – ML Forecasting (SageMaker Serverless):** Replaces browser-side CNN/LSTM with a Temporal Fusion Transformer (TFT) deployed on SageMaker Serverless Inference (pay-per-request), delivering 1h, 4h, and 12h river flood stage forecasts.

### 3 Monthly Infrastructure Cost Plan (USD & INR)

Table 1 presents the itemized monthly expenditure for a **50-node pilot deployment** in AWS ap-south-1 (Mumbai), showing both USD (\$) and INR amounts (**INR 84.50 / USD**). The table is formatted to fit cleanly within standard page margins.

#### 3.1 Multi-Tier Scaling Projections (50 to 2,000 Nodes)

Table 2 demonstrates economies of scale as FloodEye expands from a regional pilot to state-wide and national deployment tiers across India.

### 4 Cost Optimisation Strategies

By implementing AWS well-architected cost optimization techniques, the effective operational cost of the 50-node pilot can be reduced by **30–40%**:

1. **RDS Reserved Instances (1-Year Term):** Committing to a 1-year No-Upfront Reserved Instance for PostgreSQL reduces database compute costs by  $\approx 35\%$ , saving **\$12/month (INR 1,014/month)**.
2. **MQTT Payload Compression (CBOR):** Encoding ESP32 telemetry payloads in Concise Binary Object Representation (CBOR) instead of verbose JSON reduces MQTT byte volume by 50%, cutting network ingestion overhead.

| Pillar                                               | AWS Service           | Specification              | Unit (\$)     | USD/mo            | INR/mo          |
|------------------------------------------------------|-----------------------|----------------------------|---------------|-------------------|-----------------|
| MQTT Protocol                                        | AWS IoT Core          | 8.64M<br>msgs/month        | \$1.00 / M    | \$10.00           | 845             |
| PostgreSQL                                           | RDS db.t4g.small      | 2 vCPU, 2 GB, 50<br>GB SSD | On-Demand     | \$34.00           | 2,873           |
| Multi-Node App                                       | ECS Fargate + ALB     | 2 tasks, 0.5 vCPU<br>each  | Per-vCPU/hr   | \$28.00           | 2,366           |
| SMS Gateway                                          | SNS + Lambda + Twilio | 5k SMS + 100K in-<br>vokes | \$0.005 / SMS | \$25.00           | 2,113           |
| ML Inference                                         | SageMaker Serverless  | 100K reqs, 2 GB<br>mem     | Per-invoke    | \$15.00           | 1,268           |
| CDN / Network                                        | CloudFront + Route53  | 100 GB out, 1M<br>WAF req  | \$0.085 / GB  | \$12.00           | 1,014           |
| Cache Layer                                          | ElastiCache Redis     | t4g.micro (0.5 GB<br>RAM)  | On-Demand     | \$12.00           | 1,014           |
| Monitoring                                           | CloudWatch + X-Ray    | 10 GB logs/month           | \$0.50 / GB   | \$8.00            | 676             |
| <b>Total Monthly Cost – 50-Node Production Pilot</b> |                       |                            |               | <b>\$144.00</b>   | <b>12,168</b>   |
| <b>Annualized Infrastructure Budget (12 Months)</b>  |                       |                            |               | <b>\$1,728.00</b> | <b>1,46,016</b> |

Table 1: Itemized Monthly AWS Infrastructure Budget (ap-south-1 Mumbai, July 2025)

| Deployment Tier       | Nodes | Msgs/mo | Database Sizing               | USD/mo  | INR/mo   |
|-----------------------|-------|---------|-------------------------------|---------|----------|
| Pilot (River Basin)   | 50    | 8.64M   | db.t4g.small (2 vCPU, 2 GB)   | \$144   | 12,168   |
| District (Kamrup)     | 200   | 34.56M  | db.t4g.medium (2 vCPU, 4 GB)  | \$380   | 32,110   |
| State-Wide (Assam)    | 500   | 86.40M  | db.r7g.large (2 vCPU, 16 GB)  | \$920   | 77,740   |
| National Grid (India) | 2,000 | 345.60M | db.r7g.xlarge + Read Replicas | \$3,200 | 2,70,400 |

Table 2: Infrastructure Cost Projections by Deployment Tier (USD &amp; INR)

- 3. Seasonal SageMaker Scaling:** Monsoon season in Assam spans May to September. During dry winter months, SageMaker Serverless concurrency limits can be scaled to zero, saving **\$15/month (INR 1,268/month)** in idle inference fees.
- 4. S3 Glacier Tiered Storage:** Telemetry records older than 90 days are automatically migrated from RDS gp3 SSD (\$0.115/GB) to Amazon S3 Glacier Flexible Retrieval (\$0.004/GB), cutting historical data archiving costs by **96%**.
- 5. EC2 Spot Instances for Retraining:** Running weekly Temporal Fusion Transformer model retraining jobs on EC2 Spot Instances saves up to **70%** compared to On-Demand GPU instances.
- 6. CloudFront Edge Caching:** Caching frontend UI bundles and static GeoJSON flood maps at CloudFront edge locations reduces Fargate data transfer out by 80%, saving **\$9/month (INR 760/month)**.

**Optimized Pilot Cost:** Applying these six strategies reduces the 50-node monthly expenditure from **\$144 (INR 12,168)** down to approximately **\$95/month (INR 8,028/month)**, or an annualized budget of **INR 96,336/year**.

## 5 Phased 6-Month Implementation Roadmap

| Phase | Enhancement Pillar            | Key AWS Managed Services               | Timeline  |
|-------|-------------------------------|----------------------------------------|-----------|
| 1     | PostgreSQL Database Migration | Amazon RDS for PostgreSQL, Prisma ORM  | Month 1   |
| 2     | MQTT Mutual TLS Ingestion     | AWS IoT Core, X.509 Certs, IoT Rules   | Month 2   |
| 3     | Multi-Node Dashboard Mesh     | AWS ECS Fargate, ALB, CloudFront CDN   | Month 3   |
| 4     | SMS/WhatsApp Alert Gateway    | Amazon SNS, AWS Lambda, Twilio / MSG91 | Month 4   |
| 5     | Server-side TFT ML Model      | Amazon SageMaker, EventBridge, S3      | Month 5–6 |

Table 3: Phased Production Rollout Roadmap (Months 1 to 6)

## 6 Summary

The **FloodEye** platform can transition from an HTTP-based single-node prototype to a fault-tolerant, 50-node production flood monitoring network on AWS ap-south-1 for an unoptimized monthly budget of **\$144.00 (INR 12,168)** and an optimized budget of **\$95.00 (INR 8,028)**. This architecture guarantees sub-second MQTT telemetry, resilient time-series storage, automated multi-channel emergency broadcasting, and continuous machine learning flood stage forecasting.